

Tzu-Ching Liao

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Summary

Data Science MS candidate at USC designing LLM-orchestrated multi-model systems that route time series tasks across domains(healthcare, finance, transportation) to specialized forecasting modes.

Education

University of Southern California, Master of Science in Applied Data Science	Aug. 2025 – May. 2027
National Taipei University, Bachelor of Business Administration in Statistics	Sep. 2020 – Jun. 2024

Professional Experience

Research Intern Machine Learning and Data Mining Lab, Department of Computer Science, USC	Mar. 2026 -
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- Designed and developed an LLM-as-orchestrator framework that routes natural language task descriptions to specialized time series foundation models, applying Mixture-of-Experts-inspired dispatch logic across healthcare, transportation, finance, and weather forecasting domains.
- Contributed to the TSOchestra pipeline, integrating time series foundation models (e.g., Toto2) into a modular AI agent system that autonomously selects and coordinates forecasting models based on data characteristics and task requirements.
- Built and evaluated a traffic forecasting benchmark to validate the orchestration framework, demonstrating that LLM-driven model selection improves predictive performance over single-model baselines on real-world spatiotemporal data.

Data Engineer Intern

SYSTEX - specializing in cloud services, AI, and integrated IT ecosystems, Taipei, Taiwan

- Architected a **high-velocity SQL Lakehouse** on AWS using Glue and PySpark, integrating automated schema validation and CI/CD pipelines for S3 infrastructure management to ensure data integrity. This optimization slashed data-to-analysis **latency** by **70%**, enabling the business to shift from reactive reporting to real-time, proactive decision-making.
- Boosted executive decision efficiency by **50%** by architecting low-latency data pipelines for live Tableau dashboards, replacing static reporting with real-time interactive analytics to enable faster strategic pivots.

Undergraduate Research Assistant

Department of Statistics, National Taipei University

Jul. 2022 – Jun. 2023
Advisor: Prof. Hong-Long Wang

- Accelerated retail strategy optimization in a growing market by engineering a high-scale predictive framework (XGBoost/MLP) on **3.5 billion records**, achieving **86%** Recall for **repurchase forecasting**.
- By synthesizing **Association Analysis** and **K-Means** clustering, I transformed behavioral patterns into a 4-tier Customer Pyramid, enabling retailers to rapidly deploy personalized **CRM strategies** that maximize **customer retention**.

Project Experience

Time Series Orchestra

- Built an **ensemble of state-of-the-art time series foundation models** (Moirai, Sundial, Toto, Toto 2.0, TimesFM) with an **LLM-based commander** conceived to coordinate model weighting, using cross-validation and **SLSQP** optimization to produce probabilistic forecasts. Ranked top **2 on MASE** on GIFT-Eval, a large-scale forecasting benchmark.
- Designed a model abstraction layer separating forecasting logic from evaluation interfacing, via a common Forecaster interface and a Predictor adapter layer, allowing every foundation model and the ensemble itself to be evaluated interchangeably against the same benchmarking harness — improving extensibility for adding or swapping models.

Traffic Forecasting Benchmark

- Built a **unified benchmark** for large-scale traffic forecasting on the LargeST dataset, standardizing data processing and evaluation across classical ML, **spatiotemporal GNNs**, **State Space Models**, and **time-series foundation models** — using a **GIFT-Eval**-based pipeline that avoids constraining prediction horizons, letting every model compete on equal footing.
- Designed Mamba's selective-state SSM with a **parallel-scan algorithm** replacing sequential recurrence, improving accuracy by **23%** and cutting training time by **35%** versus the sequential baseline; further applied **in-training compression** to LRU, reducing training time by an additional **60%** with minimal accuracy loss.

Publications

- D. Cao, **T. Liao**, Y. Xu, N. Ghasem, Q. Guo, J. Liu, M. Weng, T. Hirano, and Y. Liu, "Traffic Forecasting in Practice: Comparing Graph Models and Foundation Models for Deployment-Aware Traffic Forecasting," (*under review / in submission*), 2025.